

SMART KISAN

Software Requirements Specification

for

Minor Project

**BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE &
ENGINEERING**

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SMART KISAN

1. 1. Introduction

1.1 1.1 Purpose

This Software Requirements Specification (SRS) document describes the functional and nonfunctional requirements of the **Smart Kisan**. The purpose of this document is to provide a detailed description of the system features, user interactions, and system constraints for the development of an AI-based agricultural advisory platform.

The Smart Agriculture AI System is designed to help farmers make better agricultural decisions by providing crop recommendations and detailed crop information through a webbased platform. The system uses machine learning models trained on agricultural datasets to suggest suitable crops based on environmental parameters.

This document serves as a reference for developers, project managers, testers, and stakeholders involved in the development and maintenance of the system.

1.2 1.2 Document Conventions

The following conventions are used in this document:

- **REQ-X** denotes a functional requirement.
- Requirements are grouped by system features.
- Priority levels used in this document are:
 - **High** – Essential for system operation
 - **Medium** – Important but not critical
 - **Low** – Optional or enhancement features
- Bullet points are used to represent system features and requirements clearly.

1.3 Intended Audience and Reading Suggestions

This document is intended for the following stakeholders:

Developers

To understand system requirements and implement the application.

Project Managers

To plan development stages and monitor project progress.

1.3 Testers

To design test cases based on system requirements. **End Users**

(Farmers / Agricultural Experts) To understand the capabilities of the system.

Readers should first review the **Introduction and Overall Description**, followed by **System Features** and **Nonfunctional Requirements** for a detailed understanding of the system.

1.4 1.4 Product Scope

The **Smart Agriculture AI System** is a web-based application that provides farmers with intelligent agricultural recommendations and structured crop knowledge.

The system aims to:

- Recommend suitable crops based on environmental conditions.
- Provide detailed crop information.
- Help farmers improve productivity and decision-making.
- Deliver agricultural insights through an easy-to-use interface.

The system is implemented using **Python, Machine Learning algorithms, and the Streamlit web framework.**

2. Overall Description

2.1 Product Perspective

The Smart Agriculture AI System is a **standalone web-based application** designed to assist farmers and agricultural stakeholders.

The system integrates:

- Agricultural datasets
- Machine learning prediction models
- Crop information modules
- Interactive dashboard interface

The application is developed as part of a **smart agriculture decision support system.**

System Components:

1. Data Processing Module
2. Machine Learning Model
3. Crop Information Database
4. User Interface Dashboard
5. Recommendation Engine

2.2 Product Functions

The major functions of the Smart Agriculture AI System include:

- Displaying crop information
- Providing AI-based crop recommendations
- Showing agricultural insights through an interactive dashboard
- Managing agricultural datasets
- Displaying crop cultivation details
- Providing irrigation and soil requirement information
- Allowing users to navigate crop knowledge modules

2.3 User Classes and Characteristics

The system will be used by the following user groups: **Farmers**

- Primary users of the system
- Limited technical knowledge
- Need simple interface and clear recommendations

2.4 Agricultural Researchers

- Use the system for crop analysis
- Interested in dataset insights and crop characteristics

2.5 Agricultural Students

- Use the platform for learning about crop cultivation and agricultural practices.

2.6 2.4 Operating Environment

The system operates in the following environment: Hardware:

- Desktop or laptop computer
- Minimum 4GB RAM recommended

Software:

- Python 3.x
- Streamlit framework
- Scikit-learn library
- Pandas and NumPy libraries
- Operating Systems:
 - Windows
 - Linux
 - MacOS

Browser:

- Google Chrome
- Microsoft Edge
- Mozilla Firefox

2.7 2.5 Design and Implementation Constraints

The system must follow the following constraints:

- The system must be developed using **Python programming language**.
- The web interface must be developed using **Streamlit**.
- Machine learning models must use **Scikit-learn**.
- Agricultural datasets must be used for training models.
- The system should run locally or on a web server.

2.8 2.6 User Documentation

The system will provide the following documentation:

- User Guide
- Installation Guide
- System Manual
- Project Documentation
- Online help instructions

2.7 Assumptions and Dependencies Assumptions:

- Users will have internet or local system access.
- Agricultural datasets used for training models are accurate.
- Users provide correct environmental parameters.
- Dependencies:
 - Python libraries (Pandas, NumPy, Scikit-learn)
 - Streamlit framework
 - Agricultural dataset availability

3. 3. External Interface Requirements

3.1 3.1 User Interfaces

The system will provide a graphical user interface consisting of:

Main Dashboard:

- Navigation menu
- Crop recommendation section
- Crop information modules Interface Components:
- Input forms for environmental parameters
- Navigation sidebar
- Data display panels
- Crop information pages

The interface will be designed to be **simple and farmer-friendly**.

3.2 3.2 Hardware Interfaces

The system will interact with standard computing devices including:

- Desktop computers
- Laptops

No specialized hardware is required.

3.3 3.3 Software Interfaces

The system interacts with the following software components:

Python Libraries:

- Pandas
- NumPy
- Scikit-learn
- Matplotlib Framework:
- Streamlit Dataset Sources:
- Agricultural datasets
- Crop yield datasets

3.4 Communications Interfaces The system uses:

- HTTP protocol for web interface
- Web browser interaction for user inputs

The application may be hosted on a local server or cloud platform.

4. 4. System Features

4.1 4.1 Crop Recommendation System

4.1.1 Description and Priority

The crop recommendation system suggests suitable crops based on environmental parameters such as rainfall, soil type, and temperature.

Priority: **High**

4.1.2 Stimulus / Response Sequences

User enters environmental parameters → System processes data using ML model → System displays recommended crops.

4.1.3 Functional Requirements

REQ-1

The system shall allow users to input environmental parameters.

REQ-2

The system shall process user input using a trained machine learning model. REQ-3

The system shall display recommended crops.

REQ-4

The system shall validate user inputs before prediction.

REQ-5

The system shall handle invalid input values.

4.2 4.2 Crop Information System

4.2.1 Description and Priority

This feature provides detailed information about different crops. Priority: **High**

4.2.2 Stimulus / Response Sequences

User selects crop →

System retrieves crop data →

System displays cultivation details.

4.2.3 Functional Requirements

REQ-6

The system shall store crop information.

REQ-7

The system shall display crop cultivation details.

REQ-8

The system shall provide irrigation information.

REQ-9

The system shall show soil requirements.

REQ-10

The system shall display crop benefits.

4.3 4.3 Dashboard Interface

4.3.1 Description and Priority

The dashboard provides the main navigation interface.

Priority: **Medium**

4.3.2 Functional Requirements

REQ-11

The system shall display a navigation menu.

REQ-12

The system shall allow users to select system modules.

REQ-13

The system shall display agricultural insights.

5. 5. Nonfunctional Requirements

5.1 5.1 Performance Requirements

- The system should generate crop recommendations within **3 seconds**.
 - The dashboard should load within **5 seconds**.
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5.2 5.2 Safety Requirements

- The system should prevent incorrect data processing.
 - The system should validate all user inputs.
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5.3 5.3 Security Requirements

- User data must not be stored without permission.
 - The system should prevent unauthorized access if deployed online.
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5.4 5.4 Software Quality Attributes

The system should maintain the following qualities:

Usability

The interface should be simple and easy to understand.

Reliability

The system should provide accurate predictions.

Maintainability

The system should be modular and easy to update.

Scalability

The system should allow integration of new crops and datasets.

6. 6. Project Plan

6.1 Team Members Example:

1. Vineet Patidar
2. Devansh Mishra

6.1 6.2 Division of Work

Analysis

Requirement gathering and dataset collection.

Design

System architecture, interface design.

Development

Machine learning model development and web application development.

Testing

System testing and debugging.

6.2 6.3 Time Schedule Phase Duration

Requirement Analysis 1 Week

Data Collection 1 Week

Model Training 2 Weeks

System Development 2 Weeks

Testing 1 Week

Documentation 1 Week

6.3 Appendix A: Glossary

AI – Artificial Intelligence

ML – Machine Learning

SRS – Software Requirements Specification

GUI – Graphical User Interface

6.4 Appendix B: To Be Determined List TBD-1:

Integration of weather API

TBD-2: Expansion of crop database